

Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction

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Abstract—[TODO: Write the abstract LAST, from the generated numbers: problem (a conformal guarantee calibrated on one driving dataset is deployed on another with zero labels), what was measured (gap, models, directions), the diagnosis (covariate reweighting fails; identifiability), the solution(s) that actually worked with their measured effect, and the limits. Do not state any effect that is not in generated/numbers.tex.]

Index Terms—Conformal prediction, trajectory prediction, distribution shift, uncertainty quantification, autonomous driving.

AUTONOMOUS vehicles plan around forecasts of what nearby road users will do. A forecast is only as useful as the statement it makes about its own reliability: a planner that knows a predicted region contains the true future with probability at least $1 - \alpha$ can keep clear of that region and of nothing larger. Conformal prediction (CP) supplies such a statement for any black-box predictor, with a finite-sample guarantee that requires nothing beyond exchangeability of calibration and test data [1], [2], and it has been adopted for trajectory prediction and safe planning [3]–[5].

Exchangeability is exactly what deployment breaks. A forecaster is calibrated on the data at hand—one set of cities, one sensor suite, one annotation pipeline—and then driven somewhere else. That accuracy drops when a trajectory predictor moves between datasets is documented [6], and that predictive uncertainty degrades under dataset shift is well known for classifiers [7]. What is not known is what happens to the *guarantee itself* when a conformally calibrated forecaster is moved, without any new labels, to a different real driving dataset: whether the stated $1 - \alpha$ survives, how badly it fails, and whether the standard remedies for shift repair it.

The conformal literature offers several remedies, each resting on an assumption. Weighted CP restores validity under covariate shift when the likelihood ratio is known or can be estimated [8]; more general results bound the coverage loss by a distance between calibration and test distributions [9]; online methods track shift by feeding realised errors back into the threshold [10], [11], an idea carried to trajectory prediction [5] and to robotic settings more broadly [12]; and a generative-prior approach has been proposed for planning under environment shift [13]. What these do not establish is the *static, label-free, dataset-to-dataset* setting in which a forecaster is frozen, calibrated once, and deployed: how large the failure is on real data, whether the covariate-shift remedy

suffices, and how many labelled target scenes it takes to detect or repair the failure.

Contributions: [TODO: Finalise after results. Planned, each to be kept ONLY if the pre-registered test supports it: (i) a measurement protocol and the first measurement of zero-label conformal coverage transfer between real driving datasets; (ii) a diagnosis — covariate reweighting does not repair the loss — with a proposition explaining why label-free repair is limited; (iii) the solution(s) that worked: model-uncertainty-normalised scores (H4), a label-efficient coverage audit with a closed-form label budget (H5b), a label-free shift monitor (H6); (iv) released code, pre-registration and experiment log.]

I. RELATED WORK

a) *Conformal prediction for trajectories.*: Split CP turns any predictor into one with marginal coverage guarantees [1], [2] and underlies conformal safe planning around predicted agents [3]. Multi-step and multi-modal trajectories have been handled with copula-based intervals [4], and scenario-aware intervals in a Frenet frame have been proposed on nuPlan [14] (a preprint evaluated on a single dataset). Locally weighted scores that divide the residual by an estimated spread [15] are the classical route to adaptive regions; our normalised score uses the forecaster’s own predicted scale in that role.

b) *Conformal prediction under shift.*: Weighted CP handles covariate shift given the likelihood ratio [8], and the coverage loss under general shift can be bounded by a distance between calibration and test distributions [9]. Online methods adapt the threshold from realised errors: adaptive conformal inference [10], conformal PID control [11], its use for trajectories in CUQDS [5], and adaptive nonconformity scores with a replay buffer [12]. These need a stream of ground-truth labels after deployment. Other work conditions on an observable nuisance parameter and generates synthetic data with a diffusion model [13] or treats the shift the ego vehicle itself induces in other agents [16] (both preprints). We study a different point in this space: no feedback, no generative model, and two real driving datasets.

c) *Cross-dataset trajectory prediction.*: UniTraj harmonises several driving datasets and reports that accuracy drops when models are transferred between them [6]; that study is about point accuracy, not about the validity of an uncertainty guarantee. Beyond trajectories, [7] showed that calibration of deep models degrades under dataset shift; our

question is the conformal analogue for structured, multi-modal forecasts.

II. PROBLEM SETTING

A. Conformal regions for multi-modal trajectories

A forecaster f maps a scene x to K candidate futures $\hat{y}_1, \dots, \hat{y}_K \in \mathbb{R}^{T \times 2}$ for a target agent whose true future is y . We use the nonconformity score

$$s(x, y) = \min_{k \leq K} \max_{t \leq T} \|\hat{y}_{k,t}(x) - y_t\|_2, \quad (1)$$

so that $s \leq q$ iff y lies inside the union of the K tube-shaped balls of radius q around the candidates. Given n calibration scenes from a *source* dataset, split CP sets $\hat{q} =$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, and the region $C_{\hat{q}}(x) = \{y : s(x, y) \leq \hat{q}\}$ has $\Pr[y \in C_{\hat{q}}(x)] \geq 1 - \alpha$ when the test scene is exchangeable with the calibration scenes [1], [2].

B. The transfer problem

Let P_S, P_T be the source and target scene–future distributions. The *coverage gap* of a source-calibrated threshold on the target is

$$\Delta = (1 - \alpha) - \Pr_{P_T}[s(x, y) \leq \hat{q}], \quad (2)$$

positive when the guarantee fails (under-coverage). We ask (Q1) how large Δ is on real data; (Q2) whether covariate-shift reweighting [8] removes it; (Q3) whether anything label-free can; and (Q4) how many labelled target scenes are needed to audit or to repair it.

C. What label-free repair can and cannot do

Covariate reweighting is valid when $P_T(y | x) = P_S(y | x)$. If the conditional differs—annotation pipelines, sensor noise, and agent behaviour all differ across datasets—no reweighting on x repairs the gap. More strongly, without restricting the class of shifts nothing label-free can.

Proposition 1 (Non-identifiability of a label-free threshold). *Let S_0 be a nonconformity score with a continuous distribution on the target, and consider the scale-shift family $S = cS_0$, $c \geq 1$, whose members are all indistinguishable from unlabeled target inputs. Let $\hat{q}$ be any threshold computed without target labels, so its law does not depend on c . If $\Pr[cS_0 \leq \hat{q}] \geq 1 - \alpha$ for every $c \geq 1$, then $\Pr[\hat{q} = \infty] \geq 1 - \alpha$.*

Proof. For fixed $\hat{q} < \infty$, $\mathbf{1}\{cS_0 \leq \hat{q}\} \rightarrow 0$ pointwise as $c \rightarrow \infty$. By dominated convergence, $\mathbb{E} \Pr[cS_0 \leq \hat{q} | \hat{q}] \rightarrow \Pr[\hat{q} = \infty]$, and the coverage requirement gives $\Pr[\hat{q} = \infty] \geq 1 - \alpha$. $\square$

Remark 1. *The proposition does not say label-free methods are useless; it says any guarantee for all scale shifts is vacuous, so a useful label-free method must either restrict the shift class (as our normalised score does, Sec. III-A) or be complemented by a label-based audit (Sec. III-C).*

D. Cost of using target labels

If k labelled target scenes are used directly, the resulting coverage has standard deviation about $\sqrt{\alpha(1-\alpha)/k}$ around $1 - \alpha$, so ≈ 1000 labels are needed for ± 2 points at $\alpha = 0.1$. Detecting a shortfall δ with an exact one-sided binomial test at level ℓ and power $1 - \beta$ needs

$$k \approx \left(\frac{z_{1-\ell} \sqrt{\alpha(1-\alpha)} + z_{1-\beta} \sqrt{(\alpha+\delta)(1-\alpha-\delta)}}{\delta} \right)^2. \quad (3)$$

For $\delta = 0.034$, $\alpha = 0.1$, $\ell = 0.05$, $\beta = 0.1$ this gives $k \approx 750$.

III. METHODS

A. Model-normalised score (label-free)

The forecaster emits, per mode k , a Laplace scale $(b_{k,t}^x, b_{k,t}^y)$. Let $u_k(x) = \frac{1}{T} \sum_t \sqrt{(b_{k,t}^x)^2 + (b_{k,t}^y)^2}$ and use the score $s'(x, y) = \min_k \max_t \|\hat{y}_{k,t} - y_t\| / u_k(x)$. The region is the union of balls of radius $\hat{q}' u_k(x)$. The premise is that when a shift makes scenes harder, the forecaster’s own scale widens, so the normalised score is less shift-sensitive. Its efficiency is measured by region area $\sum_k \pi(\hat{q}' u_k)^2$ at matched coverage.

B. Recalibration with k target labels

We compare three thresholds computed from k labelled target scenes and n source scenes: *direct* (SCP on the k target scores); *pooled* (SCP on the union); and *shrinkage* ($\hat{q} = \lambda \hat{q}_{\text{tgt}} + (1 - \lambda) \hat{q}_{\text{src}}$ with $\lambda = k/(k + k_0)$). We report the label budget k^* at which absolute coverage error falls below 0.02 in at least 90% of random label draws.

C. Coverage audit

Given a source-calibrated $\hat{q}$ and k labelled target scenes, an exact one-sided binomial test of H_0 : coverage $\geq 1 - \alpha$ flags a violation. We report its empirical power and its false-alarm rate on a shift-free split, alongside the theoretical budget of Eq. (3).

D. Label-free shift monitor

A domain classifier on label-free scene features (native sampling interval, nearby-agent count, map density, nearby lanes, history speed, heading change, curvature) is trained to separate source and target scenes; its AUC is the monitor. Within a dataset, the same classifier is trained per city against the rest. We test whether the AUC predicts the coverage loss (across cities, by Spearman correlation).

IV. EXPERIMENTAL SETUP

a) *Data.*: Argoverse 2 [17] and nuScenes [18], converted with ScenarioNet [19] and loaded through UniTraj [6]. Each dataset’s held-out split is divided by a fixed salted hash into a calibration half and a test half, so calibration and test scenes never overlap. The AV2 test split has no labels and is not used.

b) *Forecasters.*: AutoBot-Ego [20] (1.5M parameters), $K = 6$ modes. Because a 5 GB GPU shared with other work was not available, models were trained on CPU by fine-tuning with a decaying learning rate; all models are small and we state this as a limitation. A model zoo of independently trained checkpoints (**[TODO: fill from results: names, epochs, minADE]**) is used to test robustness of every conclusion.

c) *Protocol.*: $\alpha = 0.1$. Forward direction: calibrate on AV2 calibration half, test on AV2 test half (same-domain control) and on nuScenes (cross-dataset). Reverse direction: the swap. Confidence intervals are percentile bootstraps over scenes. Hypotheses H1–H9 and their pass thresholds were written before the corresponding runs (notes/falsification.md); those defined after seeing Run 1 are marked post-hoc there and are confirmed only on Run 2 data.

d) *Metrics.*: Coverage, coverage gap Δ , region area proxy, label budget k^* , audit power/false-alarm, monitor AUC and Spearman correlation.

V. RESULTS

[TODO: Generated from results/run2/*.json by make_results.py. Subsections, each reporting the pre-registered threshold and pass/fail: H1 gap size and CI; H2/H3 reweighting; H4 normalised score; H5 label-budget curves; H5b audit power; H6/H7 monitor and cities; H8 model zoo; H9 reverse direction. Negative outcomes are reported with equal prominence.]

VI. DISCUSSION AND LIMITATIONS

[TODO: Discuss after results: what the guarantee loss means for a planner; when the normalised score helps and when it does not; practical protocol (audit, then recalibrate).]

a) *Limitations (fixed).*: (i) Two datasets and one architecture family; (ii) small CPU-trained models, so absolute accuracy is below published state of the art; (iii) the observed gap is an association—its cause (annotation, sampling, agent mix) is not isolated, only a sampling-rate confound was tested; (iv) marginal, not conditional, coverage; (v) hypotheses added after Run 1 are labelled post-hoc; (vi) Proposition 1 covers scale shifts only.

REPRODUCIBILITY

All code, configuration files, pre-registered hypotheses, and the append-only experiment log are in the project repository (paper01-coverage-transfer). Every number and table in this paper is regenerated from the raw result files by paper/make_results.py; no result is typed by hand.

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